Cost-Floor Compression, the AI Wedge, and Civilisational Transition

A Conditional GE Assessment

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Unreleased application draft. This paper supplies a measurement contract and conditional accounting results; it does not forecast civilisational collapse, establish a realised GE substrate, or recommend a political programme.

Abstract

AI is a visible leading edge of a broader question: when the audited cost of performing specified transformations falls toward a measured feasible lower envelope, can the institutional arrangements that allocate work, income, access, and authority remain unchanged? This paper does not infer an answer from a benchmark, an API tariff, agricultural productivity, or a logistics platform. It defines an outcome-scaled cost ledger separating information, coordination, embodied, verification, governance, and externality costs. A nominally cheaper task is not thereby a cheaper successful social outcome.

The paper then states two conditional results. First, a fall in technical capability cost alone does not identify delivered adaptation, equitable access, or net benefit: it may also increase deployment, induced demand, verification load, and residual harm. Second, given a declared GE transition premise that the present civilisational regime cannot reproduce all of its defining arrangements under measured cost-floor pressure, the relevant outcome is a partition, not a prediction: either minimum viable reproduction fails (terminal discontinuity), or viable activity continues through a successor regime. The latter has an analytically open, not automatically benign, space of possibilities. LLMs are treated as a cost and deployment variable, not as a claim about agency or general intelligence. A hypothetical graph-based GE substrate is an unverified additional scenario, not evidence. The paper states data requirements, counterweights, and falsifiers for every empirical step.

1 Scope: a conditional transition question

This paper supplies a conditional classification. If a named regime cannot continue to reproduce its defining arrangements while the cost basis of its principal transformations changes, then unchanged persistence is not the relevant comparison class. The possible continuations are classified as either failure to reproduce a declared viability vector or reproduction through a materially different regime. Whether the antecedent holds, when it might hold, and what a successor would be remain open empirical and political questions.

This application extends neither the physical claims of GE nor its proposed graph-based substrate. Within the GE programme, narrative and narrativium name a way to ask whether a

collective arrangement can retain its identity while its relations, costs, and means of reproduction are reorganised. They are not substitutes for labour, productivity, price, energy, environmental, or institutional data. A GE-derived implementation efficiency gain has not been shown here. It is only a separately labelled scenario in Section 6.

1.1 Analysis lens: current manifold and possibility set

For this paper, the *current manifold* $\mathcal{M}(R_0)$ is a declared analytical state space of the present regime's reproduction arrangements, ledger variables, and measured constraints. "Manifold" is a classificatory term here, not a claim that the regime is a differentiable physical manifold. Given $G(R_0)$, the analysis maps a possibility set containing terminal discontinuity and viable non-isomorphic successor regimes. It assigns no value ordering to those members. Its output is the declared state description, conditions, and branch classification.

1.2 Continuance interpretation

A separate interpretive lens may then ask what the classification means for a specified aim. This paper records the following continuance lens: civilisational continuance—a trajectory that continues to satisfy a declared collective viability condition—ranks above terminal discontinuity. Conditional on that continuance, a trajectory ranks higher when it reduces measured human suffering S^H under a declared protocol and retains human participation P^H in the continuing system. Thus, under this lens, continuance is the best outcome; within it, lower S^H and continued P^H define the ideal direction.

This ordering is an explicit interpretive input, not a consequence of GE, the cost ledger, or the transition partition. It requires its own definitions of civilisation, humanity, suffering, participation, population, horizon, and measurement threshold. The analytical layer remains unchanged if a different interpretive lens is supplied.

The empirical object is not "civilisation" at one unmeasurable speed. A first test must name a unit u, outcome o, population, time interval, viability threshold, and allocation boundary: for example, a public service's ability to deliver a specified safe outcome, an agricultural supply system's ability to meet a nutritional standard, or a jurisdiction's ability to maintain a critical infrastructure service. A claim about civilisation requires an explicit aggregation argument; this paper supplies none.

2 Cost-floor pressure and the ledger

Definition 1 (Audited task cost)**.** *For named unit u, outcome o, task configuration q, and interval i, let*

$$C_i^{\mathrm{aud}}(q) = C_i^I + C_i^Q + C_i^B + C_i^V + C_i^G + C_i^E$$

be the cost required to produce a successful, accessible, and audited outcome. The terms are respectively information/inference, coordination, embodied material and energy, verification and repair, governance/authorisation, and allocated externality costs. Each term requires a stated payer, location, time horizon, allocation rule, and uncertainty range. A list API price is at most one component of C_i^I.

Definition 2 (Feasible lower envelope and compression)**.** *Let $F_{u,o,i}$ be the lowest observed, boundary-harmonised audited cost among configurations meeting fixed outcome, reliability, access and externality conditions. Envelope compression means $F_{u,o,i+1} < F_{u,o,i}$ for declared intervals on the*

same outcome scale; report the rate $(F_{u,o,i+1} - F_{u,o,i})/(t_{i+1} - t_i)$*. The separate gap* $C_i^{\mathrm{aud}}(q) - F_{u,o,i}$ *measures a configuration's convergence to the frontier. It can rise when the frontier improves and can be zero while all costs rise. Neither quantity is a thermodynamic floor.*

The empirical envelope differs from the logical-erasure floor discussed in Graphic Equalisation's Remark 7.18. A declining observed price does not compress Landauer's thermal bound, nor does it measure total civilisational adaptation cost.

For a named partial ledger, take the AI Index's GPT-3.5-level benchmark threshold, a million-token inference-price unit, its listed qualifying configurations and the November 2022–October 2024 interval. The reported price moves from \$20 to \$0.07 per million tokens (Stanford HAI, 2025). This instantiates an inference-price frontier only: the operator's full coordination, embodied, verification, governance and externality costs are unmeasured. The price-compression claim fails if a re-audit of the same threshold and token unit does not preserve the fall; the full audited-cost compression claim is not established by this partial ledger.

Remark 3 (Grammar-relative shapes, floors, and MDL)**.** *Here a declared outcome* o *is a shape, or a shape trajectory, rather than an invariant that a transformation must leave unchanged. A grammar* γ *is a declared repertoire of generative relations/operations; it makes a shape space* Σ_γ *available and induces a grammar-relative lower envelope* $F^{\gamma}_{u,o,i}$*. A new grammar* γ' *can change the available shape space as well as the envelope. A grammar step is recorded through a declared shape comparison map* $\Pi_{\gamma\to\gamma'}$ *when*

$$F^{\gamma'}_{u,\Pi_{\gamma\to\gamma'}(o),i} < F^{\gamma}_{u,o,i}.$$

It is thus a step change in what shape can be generated and at what audited cost, not preservation of a fixed outcome invariant.

Minimum description length (MDL) selects a sufficient account by minimising

$$L(M) + L(D \mid M),$$

the description of a model plus the residual description of its data (Rissanen, 1978). Under the GE reading used here, MDL pressure and cost-floor pressure are the same compression pressure at different realisation levels. MDL acts on an account and its generative grammar; the audited ledger records the information, coordination, embodied, verification, authorisation, and externality transformations through which shapes are realised.

This is not an equality $C^{\mathrm{aud}} = L(M) + L(D \mid M)$*. The correspondence requires a declared map from descriptive grammar to realised grammar, and every cost component remains boundary-, substrate-, and frame-relative. In particular, MDL alone does not establish an API price, energy use, total social cost, an empirical lower envelope, a grammar step, or a civilisational transition. Paper 01 treats the agreement between its constitutive MDL reading and the account as a consistency condition rather than independent evidence; this application retains that limit.*

The definition prevents a common equivocation. A system may reduce token price, labour hours, or local cash expenditure while transferring work to users, future maintenance, a less powerful population, an energy system, or an environmental sink. Such a change has not established lower audited task cost. Conversely, some costs can fall sharply without approaching a physical limit. The relevant question is which declared component falls and what remains.

For a declared outcome, let D_i be changed demand, A_i verified delivered adaptation, H_i induced demand and residual harm, and B_i recorded backlog:

$$B_{i+1} = \max\{0, B_i + \Delta D_i - A_i + H_i\}. \tag{1}$$

Let T_i denote task-relevant technical performance, X_i deployment exposure, Q_i coordination quality, and L_i response lag. Cost compression is a possible input to delivered adaptation, but not its definition:

$$A_i = f(T_i, X_i, Q_i, L_i, C_i^{\text{aud}};\ \text{access, reliability, governance}). \tag{2}$$

This notation allows a reduction in one cost term to help A_i, while the same change increases X_i, ΔD_i, or H_i.

Proposition 4 (Cost is not delivered adaptation)**.** *A measured decline in C_i^I, or in any other one component of C_i^{aud}, does not identify the sign of $\Delta D_i - A_i + H_i$.*

Proof. Equation (2) leaves access, reliability, governance, coordination, and the other cost components unspecified. Hold the observed component fixed and construct one admissible case in which lower cost enables verified service and raises A_i, and another in which deployment grows but failures, fraud, rework, energy use, or exclusion raise H_i or ΔD_i. Both cases share the observed cost decline and give different signs in (1). Therefore the cost decline alone identifies neither net adaptation nor backlog direction. □ □

Proposition 5 (Conditional backlog growth)**.** *Fix a stable common outcome scale and intervals $i = 1, \ldots, n$. If*

$$\Delta D_i - A_i + H_i \geq \epsilon > 0$$

in every interval, with no unrecorded reserve, substitution, or threshold redefinition, then $B_n \geq B_0 + n\epsilon$.

Proof. The stated positive increment makes the lower clamp in (1) inactive as a lower bound. Thus $B_{i+1} \geq B_i + \epsilon$; iteration proves the result. □ □

A converse catch-up condition remains equally important. If, while $B_i > 0$, $A_i - \Delta D_i - H_i \geq \epsilon > 0$, then $B_{i+1} \leq \max\{0, B_i - \epsilon\}$; backlog falls by at least $\min\{\epsilon, B_i\}$ in that interval. Cost compression cannot remove this recovery possibility by rhetoric.

3 What the evidence supports—and does not

The available evidence supports changes in particular task costs and production structures, not a universal cost-collapse law. The 2025 AI Index reports that inference cost at a GPT-3.5-comparable performance level fell by more than a factor of 280 from November 2022 to October 2024, alongside large advances on named technical benchmarks (Stanford HAI, 2025). That is strong evidence for a rapidly moving cost/performance frontier. It is not an all-in agent cost, a reliability guarantee, a capability measure for collective judgement, or a prediction of employment outcomes.

The agricultural record illustrates the broader, slower pattern without making AI its cause. Agriculture is the clearest historical example of a primary activity whose labour requirement and input composition have been transformed: in historical series for France, the United Kingdom, Italy, and the Netherlands, large agricultural employment shares are documented before industrialisation (Our World in Data, 2026). This is country evidence, not a reconstructed share for every human population or a single global historical estimate. USDA accounting further reports that U.S. farm output nearly tripled from 1948–2023, with aggregate inputs changing little; land and labour inputs fell while equipment and intermediate inputs grew (USDA ERS, 2026). The appropriate inference is labour, substitution, and productivity change under particular historical conditions, not a reconstructed all-in economic-cost series, costless food, or the disappearance of land, energy, water,

biology, capital, and distribution. For logistics, OECD analysis finds that digital connectivity and trade-facilitation measures such as automated border processes can be especially trade-enhancing for parcels, while regulatory and physical frictions remain material (OECD, 2021, 2025). A platform's convenience therefore cannot stand as evidence that commerce or transport is near a universal floor.

The proposed evidence protocol is task-level. It requires repeated performance-normalised cost data; the full audit boundary in Definition 1; reliability, appeal, and repair rates; accessibility and distribution; energy, water, material, and local infrastructure effects; and a comparison with an unchanged or alternative configuration. A named LLM release is admissible only with model/version, evaluation, prompt/tool setting, throughput, cache and context assumptions, tariff time, and oversight. Announced or vendor-listed pricing is a commercial input, not realised social cost.

4 Economic reorganisation is a hypothesis to test

The civilisational concern is not that lower cost automatically makes people poorer. It is that many current arrangements connect access, purchasing power, and social status to paid work under conditions of costly production. If cost-floor pressure affects a large enough set of tasks, the relationships between labour income, ownership of productive assets, market demand, and legitimacy become empirical variables rather than stable background conditions.

Regime existence and good governance are distinct. A regime can persist, extract, coerce, exclude, or externalise harms without equitable access, legitimacy, due process, or effective governance. None of those desirable properties is an antecedent of the terminal/successor partition. They enter this paper only when a named outcome claims them, or when their absence changes the declared viability vector, reliability, cost boundary, or harm ledger. A successor can therefore exist without being good; the partition does not endorse it.

The ILO finds that roughly one quarter of global employment is potentially exposed to generative AI, while judging task transformation more likely than replacement in most cases (ILO, 2023). Its later evidence records a 1.6 percentage-point decline in the global labour-income share from 2004–2024 and identifies several structural influences, including capital-augmenting technology and worker bargaining power (ILO, 2025). Neither result proves that AI causes a future income collapse. Together they justify measuring who owns productive capacity, who receives gains, whether demand remains adequate for the declared outcome, and whether institutions provide non-wage routes to access and agency when those outcomes are at issue.

Competition and legitimacy are optional evaluative and pathway variables, not requirements for mere system continuation. The UK CMA has identified risks to fair, open, and effective competition in foundation-model markets (CMA, 2024). The IEA reports that social acceptability of energy-intensive data-centre projects can depend on demonstrable local benefits and fair allocation of costs (IEA, 2026). These are evidence relevant to named equitable or accepted deployment pathways, not conditions a system must meet in order to persist.

Remark 6 (A testable economic transition claim)**.** *For a named population, the economically relevant claim is not "the economic system is destroyed." It is: after a declared cost reduction, do verified access, income security, agency, and outcome delivery keep pace with changes in labour demand, asset ownership, prices, and environmental burden? A negative result for one population or sector does not establish a civilisational result; a positive result falsifies a claim of necessary local exclusion. A system can continue despite either result; the result instead classifies the named quality of continuation.*

5 The GE-conditioned fork

Definition 7 (Regime and transition premise)**.** *Let R_0 be a declared present regime, represented by a set of defining reproduction arrangements $\mathcal{I}_0$ and a minimum viability vector v. The* GE transition premise *$G(R_0)$ is the additional theoretical and empirical premise that, under the stated cost-floor pressure, R_0 cannot continue indefinitely while retaining all of $\mathcal{I}_0$. This paper does not establish $G(R_0)$.*

A terminal discontinuity *occurs when the declared viable reproduction condition v fails. A* successor regime *occurs when viable reproduction continues but the arrangement is no longer isomorphic to $\mathcal{I}_0$ under a stated identity test. The successor may be more, less, or differently inclusive, stable, sustainable, or free.*

Proposition 8 (Conditional transition partition)**.** *Assume $G(R_0)$ and an exhaustive, mutually exclusive viability classification: a post-transition trajectory either fails v or continues to satisfy v. Then every classified trajectory is either a terminal discontinuity or a successor regime.*

Proof. By the viability classification, a trajectory fails v or satisfies v. In the first case it meets the definition of terminal discontinuity. In the second, $G(R_0)$ excludes indefinite retention of every defining arrangement in $\mathcal{I}_0$, so it meets the definition of successor regime. The cases are mutually exclusive by the classification. □ □

This is a partition conditional on $G(R_0)$, not a date, probability, or prediction. It gives no cardinal measure of the successor's possibility space. It means only that a viable successor is not constrained to preserve the present regime's full identity. A broad description such as "the end of the Anthropocene," "phase change," "singularity," or "Lyapunov-horizon collapse" is admissible only as a declared analytical umbrella for this question. It is not a formal geological epoch claim, demonstrated physical singularity, known mechanism, or end of human influence or geology.

6 Human timescales, physical constraints, and the GE scenario

Human biology is one constrained and socially mediated timescale among learning, care, cohort replacement, organisation, law, infrastructure, and Earth-system processes. This distinguishes the fixed individual-budget premise in *Adaptability*'s Remark 13.1 from collective adaptation: partitioning, tools and organisational changes can change the effective bound. Neither a universal institutional rate nor its negation follows from biological development alone. Development and skill formation are responsive to social conditions and cumulative investment (NASEM, 2019; Heckman, 2006); institutions can coordinate under specified conditions, and can also fail through fragmented authority, missing resources, and illegitimacy (Ostrom, 1990; IPCC, 2022).

Cost compression does not abolish physical conditions. IEA analysis reports both possible operational gains from AI and rebound risks, and projects growing data-centre electricity demand in its scenarios (IEA, 2025). Its 2026 update stresses that per-task efficiency can fall while total demand rises with volume and more intensive workloads (IEA, 2026). Every proposed floor must therefore report the energy, material, water, land, grid, labour, and local burden boundary.

A realised GE graph substrate would enter the ledger only if an independently reproducible implementation showed a lower C^{aud} at the same pre-registered task, reliability, access, and externality conditions. Until then, it is an unmeasured scenario: it may lower a technical component, leave the total unchanged, or introduce new costs. It cannot be used to establish the transition premise or accelerate a forecast.

7 Falsifiers and conclusion

The cost-floor thesis fails for a named case if: (i) performance-normalised, boundary-harmonised lower envelope does not decline on the fixed outcome scale; (ii) the apparent saving is a transfer to an omitted payer, population, time, or environmental sink; (iii) verified adaptation, access, and repair do not improve despite the saving; or (iv) a comparable configuration provides the outcome at lower total cost. The economic reorganisation hypothesis fails locally if measured access, income security, agency, demand, and legitimacy remain adequate under the declared task transformation. The transition proposition does not apply unless $G(R_0)$ and the viability classification are made explicit and independently contestable.

The evidence supports neither an inevitable collapse nor a guaranteed post-scarcity successor. It supports a more disciplined question. Rapid falls in selected information and coordination costs, together with longer-run productivity and automation changes, can place pressure on arrangements built around cost, work, and ownership. Whether that pressure becomes backlog, adaptation, terminal discontinuity, or a viable successor regime is a matter for declared measurements, not a title-level prophecy.

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